AI-101 MINI PROJECT · TEAM 75

Prediction of Price of EV

Estimating an electric car’s price from its battery, range and performance specifications

Repository github.com/praveenya1374/ai-project-team-75

THE TEAM

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The problem and the data

What we predict

Our dataset lists 10,000 EVs with their specifications and price. We train a model to estimate an EV's price in INR Lakh from those specifications.

Why it matters

Price depends on battery, range, power and more at the same time. The model shows how much each specification adds to the price.

10,000
rows

9
columns

8
input features

0
missing values

Dataset india_ev_car_features_price.csv (AI-generated dataset)
Inputs Battery capacity, range, motor power, top speed, charging time, seats, fast-charging support, boot space
Target Price (INR Lakh)

Our approach

1

Load data

Read the EV CSV with pandas; inspect shape and types

→

2

Clean

Check for missing values; median fill if any (none found)

→

3

Features

8 specifications as X, price as y

→

4

Train

80/20 split. Multiple Linear Regression on all 8 features

→

5

Evaluate

Predict on 2,000 test cars; MAE, MSE, RMSE, R^2

Libraries used

pandas
Data handling

NumPy
Numerical work

Matplotlib
Charts

scikit-learn
Model and metrics

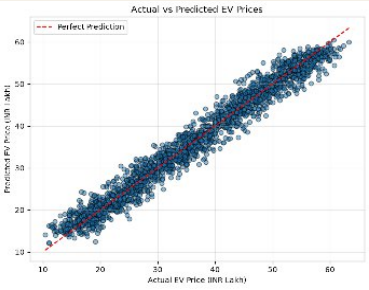
Results

0.9665 R^2 score

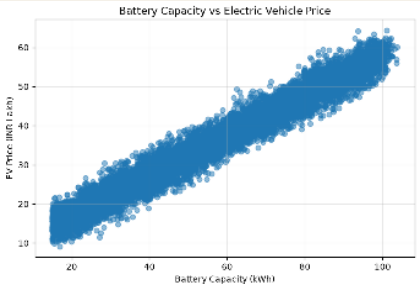
1.908 MAE (Lakh)

2.417 RMSE (Lakh)

5.842 MSE



Actual vs predicted (test set)



Battery capacity vs price

Feature	Coef.	Feature	Coef.
Battery (kWh)	0.4136	Range (km)	0.0073
Motor power (kW)	0.0228	Top speed (km/h)	0.0110
Charging time (h)	0.0094	Seats	0.2057
Fast charging	1.2949	Boot space (L)	0.0042
Intercept	1.6086		

In [ ]:

```
# 14.2 Predicting the estimated price

predicted_price = ev_model.predict(new_ev)

print(
    "Estimated Electric Vehicle Price:",
    round(predicted_price[0], 2),
    "INR Lakh"
)
```

```
Estimated Electric Vehicle Price: 38.24 INR Lakh
```

Demo: predicting the price of a new EV (cell 14.2)

Inputs: 60 kWh, 400 km, 150 kW, 150 km/h, 4 h charge, 5 seats, fast charging, 350 L boot

Learnings

What worked

- Plain linear regression reached R^2 0.9665
- Battery capacity tracks price almost linearly
- Each kWh of battery adds about 0.41 Lakh; fast charging adds about 1.29 Lakh
- Clean data: no missing values to fix

What didn't

- One 80/20 split, no cross-validation
- Only a linear model; no other models compared
- Dataset is AI-generated: the high R^2 may partly reflect its near-linear pattern, so real prices may differ
- Next: try Random Forest and cross-validate

Who did what

- Praveenya: dataset loading, exploration, cleaning
- Sriram: features, target, EDA charts
- Aayan: split, training, evaluation
- Ashok: final plots, new-EV prediction, integration, docs, repo

AI-use disclosure Our dataset is AI-generated. AI was used only for README formatting and project code structuring. No AI was used to generate the results.